

An MCM Paper Made by Team 2309944

Summary

With the increasing global attention on environmental protection, the traditional GDP, which cannot measure the environmental costs of economic development, is considered as an imperfect indicator of economic development. Some countries may achieve high-speed GDP growth at the cost of environmental destruction in the short term, but such development is unsustainable in the long run. Therefore, the GGDP, which can be used to measure environmental costs, has been created. However, currently, few countries use this indicator as their primary economic measurement method. To further protect our ecological environment, strengthen the environmental awareness of all countries, and convince more countries to use GGDP as their primary economic measurement indicator deserves our attention.

Firstly, we improved the existing GGDP calculation model and constructed a model to calculate the GGDP. In this model, we divided GGDP into traditional GDP and ecological GDP. When calculating ecological GDP, we consider both the negative impacts of human activities and the value of ecological service index(ESI).When calculating the ecological cost by human activities ,we included the pollution cost of greenhouse gases, allowing our model to reflect the impact of climate mitigation effect. The ESI only including the value can be considered as a final product to human.

Next, we constructed a nonlinear programming model to find the optimal value of GGDP and the behaviors that a government should take to achieve this goal. Because the CO₂e emissions depend on the area of a country's forest, wetland, grassland, cropland, and lake, we used the annual change rate of these five types of land as the decision variable to determine how a country should change its land use to achieve the optimal GGDP.

To analyze the expected impact of changes in economic measurement indicators on the global environment, we selected 14 typical countries distributed globally with different levels of economic development for analysis, and used these countries to represent the global impact. Applying the optimization model to the samples of these 14 countries from 2012 to 2020 for 8 years, we obtained the corresponding optimal GDDP and land change rate for each year. Then, based on this change rate, we predicted the reduction of CO₂e emissions from 2021 to 2030 and illustrates its global effect. To illustrate the rationale for a country to switch its economic measurement indicator, we conducted a beneficial and cost analysis using the expected reduction in CO₂e emissions and the opportunity cost of protecting the environment.

Finally, based on the optimal results of the 14 countries mentioned above, we found that China has great potential for improvement. So we conducted a further in-depth analysis of it. The results showed that during the early stages of GGDP policy, China's economic development will be impacted to some extent but will gradually recover to the actual level. The ability to provide for future generations will also significantly improve. Overall, considering long-term sustainable development, China should adopt GGDP as its primary economic measurement indicator. We also wrote a report recommending the implementation of this policy to the Chinese government.

Additionally, we conducted a sensitivity analysis of the model and summarized the strengths and weaknesses of our work.

Keywords: Environmental Protection, Climate Mitigation, Greenhouse Gases, GGDP, Sustainable Development, Nonlinear Programming Model

Contents

1	Introduction	2
1.1	Problem Background	2
1.2	Our Work	2
2	Assumptions and Justifications	3
3	Notations	4
4	Model 1:GGDP Valuation	5
4.1	Basic Valuation Method	5
4.2	Computing Ecological Service Index	6
4.3	Computing Ecological Cost	7
5	Model 2: Expected Behavior After Adopting GGDP	8
5.1	Formulate the Objective Function	8
5.2	Define the Decision Variables	9
5.3	Specify the Constraints	9
5.4	Find the Optimal Solution	10
5.4.1	Interior Point Method	10
5.4.2	Sequential Quadratic Programming	10
5.4.3	Active Set Method	10
5.4.4	Trust Region Reflective Method	10
5.5	Why defendable?	11
6	Global Scale Climate Impact	11
6.1	Typical Countries Selection	11
6.2	Expected Climate Impact	12
6.3	Benefit-Cost Analysis	15
7	Deep Analysis in China	17
7.1	Use of Natural Resources	17
7.2	Net Benefit	17
7.3	Change of Economic Status	19
7.4	Improvement of Ability to Support Generation	19
7.5	Sensitivity Analysis	19
8	Evaluation of Models	20
8.1	Strength	20
8.2	Weaknesses	21
9	Conclusion	21
	Non-technical Report	22
	References	23

1 Introduction

1.1 Problem Background

Since its proposal in 1937, GDP has been used to measure national economic activity and evaluate policy decisions and the state of the national economy. Traditional GDP is defined as "the monetary value of final goods and services produced in a country in a given time period; it counts all of the output generated within the borders of a country" (Callan, T., 2023). However, traditional GDP fails to account for the unobservable part of economic activity.

The first imperfection of the GDP indicator is its inability to capture the sustainability of economic growth, as it fails to account for the depletion of natural resources that can result from excessive exploitation (Rauch and Chi, 2010). A secondary imperfection is its inherent limitations in serving as a reliable measure of social welfare, as it fails to capture important factors such as volunteerism, the cost of environmental degradation, leisure time, and others. Therefore, to accurately evaluate the real impact of economic growth and ultimately gauge the sustainability of an economy, it is imperative to utilize some form of environmental accounting (Li and Lang, 2010).

The British economist Wulfde sharply pointed out that if a country only increases its material capital while reducing its environmental capital, the overall capital may be zero or even negative, and development becomes unsustainable. In pursuit of their own interests, humans have blindly exploited and damaged the natural environment, causing environmental destruction, excessive greenhouse gas emissions resulting in the greenhouse effect, global sea level rise, and the submergence of some islands, leading to a large number of refugees. Extreme weather events have increased, biodiversity has decreased, viruses have spread, and human survival is seriously threatened. Fortunately, humans have transitioned from the primitive concept of human triumphing over nature to the sustainable development concept of harmonious coexistence with nature. In 1987, the Brundtland Commission submitted a report to the United Nations entitled "Our Common Future," which explicitly proposed the concept of sustainable development—"Sustainable development means meeting the needs of the present without compromising the ability of future generations to meet their own needs."

To address the limitations of GDP as a measure of economic development, the concept of Green GDP has emerged. Green GDP is a general concept that encompasses various adjusted GDP measures that factor in social and environmental costs, including those that are not traditionally expressed in monetary units. International organizations such as the United Nations and the World Bank have been conducting extensive research and promoting the use of Green GDP since the 1970s. By advocating for the use of Green GDP to measure economic development, countries can prevent blind pursuit of short-term, high-speed GDP growth at the cost of environmental damage and excessive use of resources. Incorporating social and ecological costs is intended to impose penalties on countries that achieve development at the expense of the environment, thereby making all countries aware of the importance of protecting the environment and fostering a sustainable development system.

1.2 Our Work

In order to promote countries in the world to use GGDP to measure their development and reach the sustainable developing goal, we have completed the following tasks:

- First, we construct a green GDP calculation method based on relevant literature to mea-

sure both traditional GDP and the environmental costs associated with economic development.

- Next, we create a nonlinear programming model with GGDP as the objective function and the percentage change of each type of land in a country as decision variables, subject to constraints. We assume that all countries want to maximize their GGDP.
- Furthermore, we measure the impact of country behavior on climate improvement and the associated costs after adopting GGDP. We consider using GGDP to measure economic development worthwhile as long as the benefit-cost ratio gradually increases and remains above 1 in the long term.
- Additionally, to verify the model's feasibility, we select China as a typical case and study the changes in the country's various ecosystem areas and economic development levels after replacing GDP with GGDP. We analyze whether GGDP can promote a country's long-term development and help it achieve sustainable development goals.
- Finally, based on our analysis of this typical case, we reach a conclusion on whether GGDP should replace GDP and write a non-technical report for the Chinese government explaining the advantages and disadvantages of GGDP and helping them choose an indicator to measure economic development.

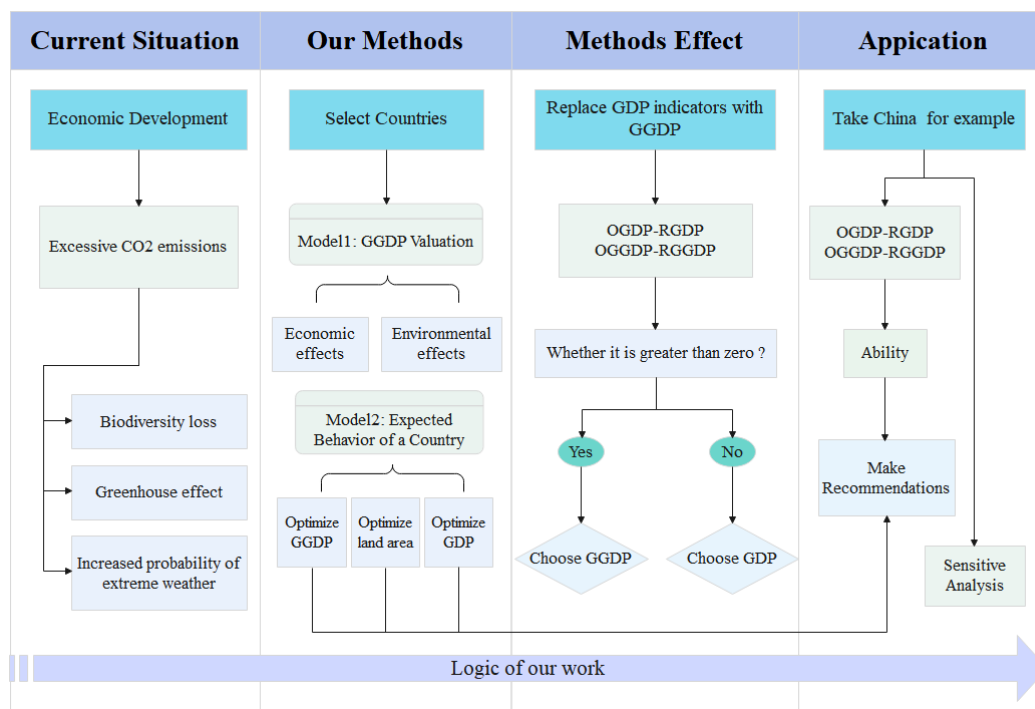


Figure 1: Logic of our work

2 Assumptions and Justifications

- Assuming that all countries make rational and independent decisions, their goal after using GGDP as an alternative economic measure would be to optimize their GGDP.

- Assuming CO_2e (carbon dioxide equivalent) covers all greenhouse gases emitted by a country, different greenhouse gases such as CH_4 and N_2O are assigned weights based on their Global Warming Potential (GWP) and expressed in a standardized unit.
- Assuming ESI is independent of the emission or absorption of greenhouse gases, counted at fine spatial and temporal scale. Ecological Service Index (ESI) refers to the value that natural resources provide to humans. To facilitate the estimation of the impact of using green GDP on the climate, we extract the net value of natural resources' impact on greenhouse gases (positive or negative) from the original value of natural resources.
- Assuming that in the short to medium term, the total rate of reasonable land use changes in a country cannot reach its limit. There are limits to changes in the area of each type of land in a country, and it cannot continue to expand without limits. However, this limit is only likely to be reached in the long term.
- Assuming that for the same country, the pattern of natural resource use after implementing the policy of using GGDP as the primary economic measurement indicator remains similar, regardless of when the policy is implemented. So we can apply the natural growth rate obtained based on using GGDP as the primary economic measurement indicator starting from 2012 to any year in the future where policy changes occur. Because carbon can be measured in terms of price in 2012 when the opening of carbon market trading.

3 Notations

The primary notations used in this paper are listed in Table 1.

Table 1: Notations

Symbol	Definition
$GGDP$	The green GDP
ESI	Ecosystem Services Index, measuring ecosystem service values at a final spatial and temporal scale
ESV	Ecosystem Services Value, measuring total ecosystem service values
$Ability$	Ability to support generation, $Oability$ the ability to support generation after adopting GGDP, $Rability$ is the corresponding indicators based on our real worlds.
$UCO2e_k$	The carbon dioxide emissions per unit area of the k-th ecological system
V_k	Ecosystem service value of per unit area of k-th ecosystem, considering only the portions of ecosystem services that can be regarded as final products
TV_k	Total ecosystem service value of per unit area of k-th ecosystem
A_k	The area of k-th ecological system, OA_k the optimal area of the k-th ecosystem in the country after adopting GGDP as the primary economic index, RA is the actual area of the k-th ecosystem in this country
k-th	The k-th ecological system includes forest, grassland, wetland and so on

4 Model 1:GGDP Valuation

4.1 Basic Valuation Method

Currently, there are two main types of approaches for measuring Green GDP (GGDP) as discussed by Yue and Xu (2008, 2009). Type I GGDP accounts for GDP minus the cost of environmental pollution and resource depletion, but it neglects the value of natural ecosystem services. On the other hand, Type II GGDP was first proposed by Costanza (1997), who estimated the global value of ecosystem services and introduced the concept of including eco-service valuation in the traditional economic accounting system (Costanza, 2000). Boyd and Banzhaf (2007) noted that being valuable and being a service are not equivalent. The valuation of ecosystem services is dependent on their spatial and temporal demand, which varies greatly. Thus, ecosystem services must be accurately assessed at a fine spatial and temporal scale.

Gao (2005) proposed a method of green GDP accounting for agricultural production based on the ecosystem services provided by the agricultural system. This approach considers both the value of ecosystem services and the ecological cost associated with agricultural production. We refer to this approach as Type III green GDP. Type I green GDP only takes into account the pollution and resource depletion caused by human activities, while ignoring the long-term value of natural resources, such as for culture and recreation. Type II green GDP considers the value of natural ecosystem services, but it is difficult to account for the damage caused by some human activities, such as greenhouse gas emissions. In a sense, the value of natural ecosystem services can be used to measure the value of natural resources for greenhouse gas purification,

but the pollution caused by human activities (such as industrial production) should also be subtracted. Therefore, we adopt Type III green GDP, which measures both the pollution and damage caused by human activities and the value of natural resources for providing ecosystem services to humans.

Referring to Gao et al. (2005), Stjepanović (2017), and Costanza et al. (2014), we use Type III green GDP in this paper and attempt to construct a GGDP calculation framework that can be applied and compared across countries. The GGDP of a country j can be calculated as equation (1) and (2):

$$GGDP_j = TraditionalGDP_j + EcologicalGDP_j \quad (1)$$

$$EcologicalGDP_j = ESI_j - EcologicalCost_j \quad (2)$$

The ESI refers to the Ecological Service Index which measures the economic value of the final ecosystem services provided. The detailed calculation methodology is provided in Section 4.2 and Section 4.3.

4.2 Computing Ecological Service Index

Ecosystem Services Index(ESI) represents the value of final ecosystem services. As the kind of ecosystem services depends on where and when the demand for, it must be counted at fine spatial and temporal scales (Boyd and Banzhaf, 2007).

Ecological service value(ESV) is a concept that measures the benefits that ecosystems and their components, such as plants and animals, provide to humans. These benefits can be categorized into four types: provisioning services, regulating services, cultural services, and supporting services. A more comprehensive and specific classification can be found in 2.

When calculating green GDP, it is essential to account for ecosystem services at a final spatial and temporal scale, as value and service are not equivalent concepts (Boyd and Banzhaf, 2007). However, most studies simply add up the values of ecosystem services and GDP without clearly distinguishing their respective contributions, as was done by Gao (2005). In some cases, certain values may be double-counted, such as food supply and raw materials, as their corresponding ecosystem values may have already been reflected in GDP. To address this issue, our study, following Xu et al. (2010), seeks to derive the replacement ecological service value, which is referred to as ESI. Detailed calculation methods are described in sections 3.2 and 3.3.

Costanza et al. (2000) first proposed a method for measuring the value of ecosystem services, which includes 17 categories of ecosystem value. To more accurately identify the components of ecosystem service value that should be included in green GDP, we followed the approach of Xu et al. (2010) and considered six types of ecosystem services, namely water regulation, soil formation, waste treatment, genetic resources, culture, and recreation. It is worth noting that the value of climate regulation is not included. In the following sections, we separately consider the value of the ecosystem's impact on climate, which makes it easier to assess the impact of using GGDP on climate.

Based on the research conducted by Costanza et al. (2000) and Xu et al. (2010), we have obtained the unit value of various types of ecological systems at the global level and have calculated the final value of each system. The Ecological Service Index (ESI) measures the value of the useful components of ecosystem services, which depends on the location and timing of demand.

	Forest		Grassland	Wetland		Lake/River	Cropland	Others			
	Tropical	Temperate	Grassland	Tidal marsh	Swamp	Lakes/River	Cropland	Deseret	Ice/rock	Tundra	Urban
Gas regulation	\	\	7	\	265	\	\	\	\	\	\
Climate regulation	233	88	0	\	\	\	\	\	\	\	\
Disturbance regulation	5	\	\	1839	\	\	\	\	\	\	\
Water regulation	6	\	3	\	7240	5445	\	\	\	\	\
Water supply	8	\	\	\	30	2117	\	\	\	\	\
Erosion contrd	245	\	29	\	7600	\	\	\	\	\	\
Soil formation	10	10	1	\	\	\	\	\	\	\	\
Nutrient cycling	922	\	\	\	\	\	\	\	\	\	\
Waste treatment	87	87	87	6696	1659	666	\	\	\	\	\
Pollination	\	\	25	\	\	\	14	\	\	\	\
Biological contract	2	4	23	\	\	\	24	\	\	\	\
Habitat	\	\	\	169	439	41	\	\	\	\	\
Food production	32	50	67	466	47	\	54	\	\	\	\
Raw materials	315	25	\	162	49	\	\	\	\	\	\
Genetic resources	41	\	0	\	\	\	\	\	\	\	\
Recreation	112	36	2	685	491	230	\	\	\	\	\
Cultural	2	2			1761	\	\	\	\	\	\
Unit Value(ESV)	2007	302	232	9990	19580	8498	92	0	0	0	0
Unit Value(ESI)	258	135	93	7381	11151	6341	0	0	0	0	0
Category unite value(ESI)	183		93	9266		6341	0	0			

Figure 2: The unite value of different ecosystem (1994 \$/ha/yr)

Figure 2 shows the original unit values in 1994. The 17 kinds of service value shown in the figure2 are considered in the calculation of traditional ecosystem service values(ESV). While the deep green values refer to the values that are considered in the ESI (Ecosystem Services Index) calculation in this paper, which enables the measurement of ecosystem values at a final spatial and temporal scale, only including the parts that are unrelated to GDP calculation and climate impact are retained. In practical applications, we adjust the unit value to its current-year value using the Consumer Price Index (CPI) of the United States. In this paper, ecosystem services index is defined as follows: (3):

$$ESI_j = \sum_{k=1}^n A_{kj} \times V_k \quad (3)$$

A_{kj} represents the area of ecosystem k in country j , while V_k denotes the ecological value coefficient per unit area of type of ecosystem k . The ecological service value coefficient, denoted as V_k , quantifies the unit value of the k -th ecosystem in country j , based on the framework proposed by Costanza et al. (2000), while considering only the portions of ecosystem services that can be regarded as final products.

4.3 Computing Ecological Cost

We adopt the methodology of Stjepanović (2017) to construct a cross-national applicable framework for calculating ecological cost. In this study, the ecological cost of country j can be calculated as follows:

$$EcologicalCost_j = PCO2 \times \left(\sum_{k=1}^n A_{kj} \times UCO2e_{kj} \right) + Waste_j + \left(\frac{GNI}{100} \times NRD_j \right) \quad (4)$$

The first term represents the cost of greenhouse gas pollution, calculated as the monetary value of CO₂e emissions. CO₂e (Carbon Dioxide Equivalence) is a unit of measurement used to compare the emissions of different greenhouse gases, as each gas contributes differently to the Earth's greenhouse effect. To standardize the measurement of overall greenhouse effect, and since carbon dioxide is the primary gas generated by human activities, it is used as the fundamental unit of measurement for greenhouse effects. The carbon dioxide equivalent of a gas is

determined by multiplying its tonnage by its Global Warming Potential (GWP), which normalizes the effect of different greenhouse gases. We break down this term into net carbon dioxide emissions produced by different types of land. Specifically, A_{kj} represents the k -th ecological system area in country j , and $UCO2e_{kj}$ denotes the carbon dioxide emissions per unit area of the k -th ecological system in country j . Since the production and carbon dioxide emission capabilities differ among countries, the values of $UCO2e_{kj}$ vary across countries. Therefore, we calculate the term based on the ecological system area of each country and the corresponding CO₂e emissions for that year. The Price of Carbon Dioxide ($PCO2$) is derived from the average trading prices in the carbon markets of the United States, Europe, Japan, and China for the given year. This metric is commonly used in environmental economics to assess the economic costs of carbon emissions and to inform policy decisions related to carbon pricing and emissions reduction targets.

The second term $Waste_j$, comprises various types of waste, including domestic wastewater, incineration waste, industrial wastewater, and solid food waste. We measure the original data of these waste types using carbon dioxide equivalence. We then use the average carbon price to evaluate their economic value, resulting in the calculation of the monetary value of waste.

The final term is the adjusted savings of natural resource depletion presented as a percentage of gross national income per country. Gross national income, or GNI, is the aggregate of value added by all resident producers, in addition to any product taxes (minus subsidies) not included in output valuation, and net receipts of primary income from abroad, such as employee compensation and property income (WDI). Ultimately, the variable adjusted savings of natural resource depletion (NRD), as a percentage of GNI per country, represents natural resource depletion as a total of net forest depletion, energy depletion, and mineral depletion.

5 Model 2: Expected Behavior After Adopting GGDP

To obtain the global climate mitigation effects and global impacts contributed by a country choosing to use GGDP as its economic measure, we need to know the expected behaviors of a country to reach the maximum GGDP. Nonlinear programming is a type of mathematical optimization that deals with finding the optimal solution to a problem with a nonlinear objective function and nonlinear constraints. It can help us to optimize a real-world situation by finding the best solution given the constraints and objectives of the problem. So we can construct a nonlinear programming model to find the expected behavior of the government.

The construction steps of a nonlinear programming model include the following steps:

5.1 Formulate the Objective Function

Formulate the objective function, which is the function that needs to be optimized. If GGDP replaces GDP as the measure of a country's economy, then the objective of a country's decision-making is to optimize its GGDP. Under our assumptions, a country will accomplish its optimization objective by gradually changing its land use practices. Therefore, we can derive its optimal GGDP objective function as follows:

$$\begin{aligned} \max \quad OGGDP_j = & RGGDP_j - (\sum_{k=1}^5 \Delta A_{kj} \times UCO2e_{kj} \times PCO2_k) \\ & + (\sum_{k=1}^5 \Delta A_{kj} \times ESV_k) - GDP(CO2e) \end{aligned} \quad (5)$$

ΔA_{kj} , is the percentage change in a country's five land types, which is used to measure how the country changes its land use practices each year to achieve optimal GGDP. $GDP(CO_2e)$ is the fitted function between a country's CO_2e emissions and its GDP. It is conceivable that when a government uses GGDP as an economic measure, it will inevitably reduce the country's CO_2e emissions by stopping the cutting of trees, investing more money in the environment, reducing pasture and arable land, using land that was used for factories for environmental purposes, etc. At the same time, the change in CO_2e emissions can be directly obtained from the change in land use practices. To find the relationship between CO_2e emissions and GDP, we use the **CFTOOL** in **MATLAB** software to fit CO_2e and GDP to obtain the impact of a unit change in CO_2e on GDP, which is a measure of how a country's GDP is likely to be affected when it aims to maximize GGDP in that year. The results show the relation between GDP and CO_2e is a nonlinear function, which also indicates that our objective function is nonlinear and requires a nonlinear programming model to solve the problem.

5.2 Define the Decision Variables

Define the decision variables, which are the variables that affect the objective function. From the composition of the objective function, we can see that our decision variable is the percentage change of each type of land in a country, ΔA_{kj} , with a total of 5 indicators.

5.3 Specify the Constraints

Specify the constraints, which are the conditions that limit the possible values of the decision variables. Setting a reasonable constraint for ΔA is an important part of obtaining an efficient optimal solution from the nonlinear programming model, so we have reviewed many documents and data to ensure the validity of our constraints.

Forests, wetlands, and lakes are important components of a country's environmental system. Countries that use GGDP as a measure of economic development tend to place a greater emphasis on environmental protection. Therefore, we believe that the government of such a country will not allow significant annual decreases in the areas of these three types of land. Since the establishment of the International Day of Forests by the United Nations in 2012, countries worldwide have been urged to protect forests. As a result, we assume that the rate of forest loss since 2012 represents the minimum decline rate that governments are able to achieve despite their best efforts. Similarly, since the establishment of World Wetlands Day in 1996, the global wetland area has continued to decline by approximately 1.3% annually. Since the establishment of World Water Day in 1993, the area of lakes and rivers worldwide has decreased by around 1% per year. Furthermore, when a country optimizes its GGDP, the government is likely to allocate more funding to protect these three types of land. This should further reduce the overall decline rate.

$$\text{s.t.} \quad \begin{cases} \Delta A_{Forest} \geq -0.117\% \\ \Delta A_{Wetland} \geq -1.3\% \\ \Delta A_{lake} \geq -1\% \\ \Delta A_{Forest} + \Delta A_{Wetland} + \Delta A_{lake} \geq -2\% \end{cases} \quad (6)$$

The government of a country may find it difficult to make significant changes to the existing land use within a year, and the ability of the government to restore forests, wetlands, and lakes is also limited. Based on past data on annual growth of regenerated forests worldwide, we can estimate the maximum global capacity for forest restoration. Unfortunately, we do not have

sufficient data to measure the annual restoration of wetland and lake systems. Therefore, we have adopted an alternative approach, where we first calculate the annual rate of decline for these two types of land, and then use the reduction rate of this decline as an indicator of the global ability to restore wetlands and lakes.

$$\text{s.t. } \begin{cases} \Delta A_{\text{Forest}} \leq 1.6\% \\ \Delta A_{\text{Wetland}} \leq 0.3\% \\ \Delta A_{\text{Lake}} \leq 0.5\% \end{cases} \quad (7)$$

5.4 Find the Optimal Solution

Solve the model using optimization algorithms to find the optimal solution. Various algorithms are existed to solve nonlinear programming, including Interior Point Method, Sequential Quadratic Programming, Active Set Method, and Trust Region Reflective Method. Here is a short description of these algorithms.

5.4.1 Interior Point Method

Interior point methods use a sequence of points inside the feasible region to solve nonlinear programming problems by minimizing a logarithmic barrier function, providing a globally convergent solution. These methods are well-suited for problems with many constraints.

5.4.2 Sequential Quadratic Programming

Sequential Quadratic Programming uses a quadratic approximation of the objective function and constraints at each iteration. This approach can efficiently solve medium-sized optimization problems but may have difficulty handling nonlinear constraints, especially when the constraints are degenerate or the problem is ill-conditioned.

5.4.3 Active Set Method

The active set method selects a subset of constraints to be active and solves a linear programming subproblem to determine the search direction, effectively handling problems with a small number of constraints. However, it can be difficult when the constraints are nonlinear and the feasible region is irregular.

5.4.4 Trust Region Reflective Method

The trust region reflective method uses a trust region and reflection technique to handle constraints iteratively. This method is known for its reliability and robustness, but can be sensitive to the choice of the initial point and may require careful selection of trust region size.

After applying the four algorithms mentioned above to solve the optimization problem, we found that the Interior Point Method can achieve the best results, so we selected it as our maximizing optimization algorithm. We constructed a nonlinear programming model using Matlab to solve for the optimal GGDP and corresponding changes in land use. By multiplying the land change rates with the actual land areas, we obtained the optimized land areas, denoted as OAForest, OAWetland, OAGrass, OACropland, and OALake. Considering that changes in land use in one year will affect the actual areas of different types of land in the following year,

we dynamically adjusted the actual land areas for the next year based on the optimized land areas of the previous year when running the nonlinear programming model

$$OAForest = (1 + \Delta AForest) \times AForest \quad (8)$$

$$OAWetland = (1 + \Delta AWetland) \times AWetland \quad (9)$$

$$OAGrass = (1 + \Delta AGrassland) \times AGrass \quad (10)$$

$$OACropland = (1 + \Delta ACropland) \times ACropland \quad (11)$$

$$OALake = (1 + \Delta ALake) \times ALake \quad (12)$$

5.5 Why defendable?

Since the government needs to consider the benefits and losses of a policy before implementing it, they often rely on past data to estimate the expected outcomes. So we can assume that the government will use the past data to find methods to optimize its objectives and apply the same behaviors to future policy. Therefore, regardless of when the policy that uses GGDP as the primary economic indicator is implemented, the government's behavior is likely to be similar, and the pattern of natural resource use will be similar as well. The optimal data obtained from our model is based on past data and has strong practical and guiding significance. The optimal indicator obtained from the model includes changes in land use rather than the optimal area of each type of land, which is more suitable for the government to develop plans based on this rate of change. Therefore, we can apply the land use change rate obtained since 2012 when GGDP was used as the primary economic indicator to any year in the future when a policy change occurs to anticipate the impact of the decision on the world.

6 Global Scale Climate Impact

To evaluate the impact of using GGDP as a measure of economic performance on a global scale, we selected 14 typical countries. Firstly, assuming that green GDP became the primary measure of economic level starting from 2012, we used the model 2 to obtain the optimal values after adopting GGDP, and obtained the actual economic values based on real economic data. Besides, based on the actual and optimal values, we estimated the impact of using GGDP on global climate. We then conducted a cost-benefit analysis to determine whether the efforts and impacts on climate improvement by these typical countries are worthwhile.

6.1 Typical Countries Selection

We have selected 14 typical countries as shown in figure 4 and figure 3, including both developing and developed countries that are geographically diverse. Therefore, we believe that the conclusions based on these countries are highly credible and can be generalized to the global.

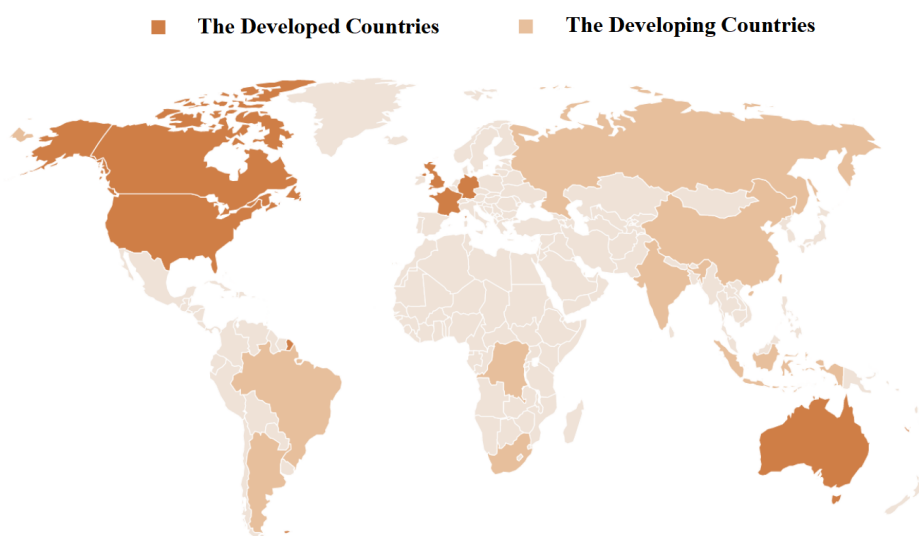


Figure 3: Distribution of typical countries



Figure 4: Introduction of Typical Countries

6.2 Expected Climate Impact

As the GGDP calculation method established in model 1, we can compute the actual GGDP (RGGDP) using observed values. By utilizing the nonlinear programming model in model 2, we are able to obtain the expected behavior of the selected typical countries, which includes the optimal GGDP (OGGDP), optimal GDP (OGDP), and optimal area of different types of ecosystems. The root cause of climate change lies in the emission of greenhouse gases, which results

in the warming of the earth's temperature. Issues such as glacier melting, extreme weather, biodiversity change, and reduced food production stem from the emission of greenhouse gases. Therefore, we attempt to calculate actual $CO_2e(RCO_2e)$ and optimal $CO_2e(OCO_2e)$. If the expected CO_2e is less than the real CO_2e , we consider that the optimal decision after adopting GGDP will lead to a reduction in greenhouse gas emissions. We measure the potential positive impact on climate in terms of the value of the reduced amount of greenhouse emission. The potential impact of a country j on climate can be measured as follows:

$$ClimateImpact_j = RCO_2e_j - OCO_2e_j \quad (13)$$

$$RCO_2e_j = \sum_{k=1}^n RA_{kj} \times UCO_2e_{kj} \quad (14)$$

$$OCO_2e_j = \sum_{k=1}^n EA_{kj} \times UCO_2e_{kj} \quad (15)$$

RCO_2e_j represents the greenhouse gas emissions when GDP is used as the economic measure, while OCO_2e_j represents the greenhouse gas emissions when GGDP is used as the economic measure. RA_{kj} denotes the area of the k -th ecosystem in country j when GDP is used as the economic measure, and EA_{kj} represents the land area when GGDP is used as the economic measure. UCO_2e_{kj} represents the unit area greenhouse gas emission of the k -th type of land in country j . We assume the value of UCO_2e_{kj} remains unchanged when calculating RCO_2e_j and OCO_2e_j , as we assume that the resource use level or carbon-saving level of a country does not change in the short term. We assume that the net emissions of CO_2e are affected by the land type and land area. For example, the net emission of each unit of forest(UCO_2e) is a negative value, which means that the greenhouse gas absorption of the forest is greater than its emission.

Figure 5 shows the changes in CO_2e emissions in these countries after implementing GGDP. We compared and analyzed the developed and developing countries separately. From the figure, it can be observed that when developed countries adopt GGDP to measure economic development, except for Australia, the CO_2e emissions of all other developed countries are lower than when GDP is used as the economic development indicator. Therefore, the CO_2 reduction values are positive, and they increase with the passage of time, with the data for the US and Canada being particularly significant. Similarly, in developing countries, all countries have reduced CO_2e emissions after adopting GGDP. The data for Russia, China, Brazil, and India show significant environmental improvements after using GGDP to measure economic development, while the data for other countries are relatively less significant.

Overall, the data from both developed and developing countries support the use of GGDP as a measure of economic development, as it significantly reduces global CO_2e emissions, thereby improving the environment and promoting sustainable human development. However, it should be noted that the lack of precise statistics on natural resource data in some countries worldwide makes it difficult to accurately assess the impact of using GGDP as an alternative indicator on global climate. Additionally, we cannot accurately predict a country's decisions after adopting GGDP as the primary economic indicator.

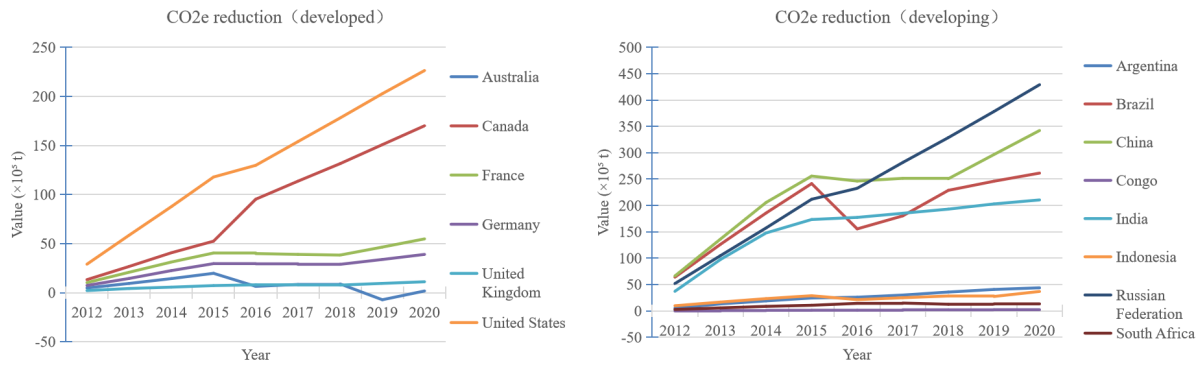


Figure 5: Change of CO2e reduction.png

Furthermore, we need to estimate the impact of this transition on future CO2e emissions. Assuming that the governments of all selected typical countries began implementing policies based on GGDP as the economic measure in 2021, as we assume that the government will engage in behavior similar to the past to achieve the optimization of GGDP, we can directly use the past optimization rate of land use change to predict the future.

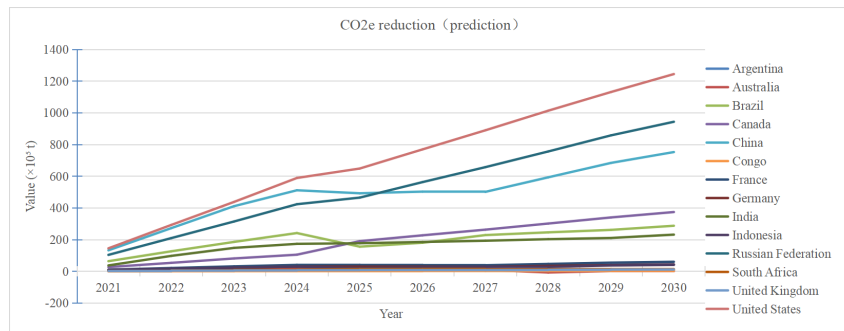


Figure 6: Prediction CO2e reduction

As shown in figure 6, all typical countries have increased their net reduction in CO2e emissions per year by optimizing their land use after implementing policies based on GGDP as an economic measure. The reduction in CO2e emissions is expected to bring multiple benefits to the global climate, as shown in the following figure:

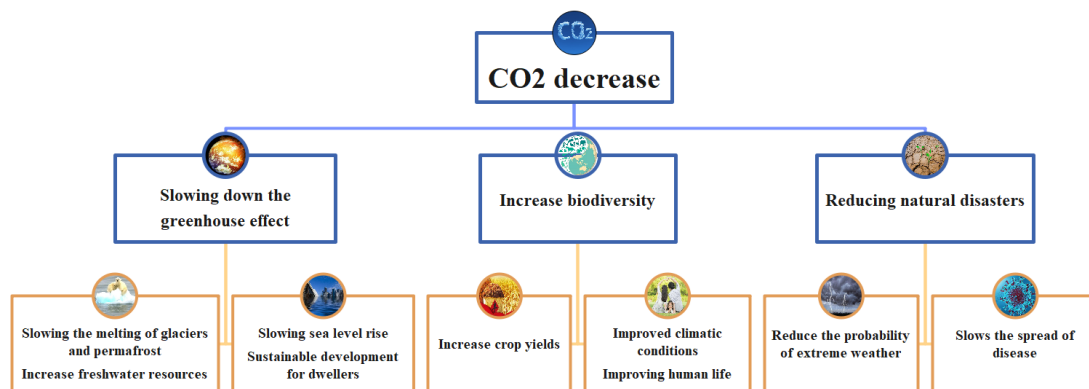


Figure 7: Impact of CO2e reduction

6.3 Benefit-Cost Analysis

Undoubtedly, using GGDP as a measurement indicator has positive impacts on the global climate, but it inevitably incurs economic costs. Therefore, we conducted a benefit-cost analysis to determine whether using GGDP as a primary measurement indicator is worthwhile.

Potential Cost

First, we estimate the required efforts to achieve the maximum GGDP, which we believe may include costs associated with promoting clean energy, implementing sustainable production and tourism, and supporting environmental research and development. However, investing in these areas may lead to a reduction in investments that stimulate traditional economic growth. In other words, the required efforts may involve some sacrifices in traditional economic growth. Therefore, we measure the required efforts as follows:

$$PotentialCost_j = OGDP_j - RGDP_j \quad (16)$$

$OGDP$ represents the expected GDP of a country after using GGDP as the measurement indicator, while $RGDP$ represents the actual GDP of the country.

Potential Benefit

We quantify the potential benefit of implementing GGDP as the monetary value of the reduction in greenhouse gas emissions.

$$PotentialBenefit_j = PCO_2 \times (RCO_{2e_j} - OCO_{2e_j}) \quad (17)$$

PCO_2 is the price of carbon dioxide, is measured as the average trading prices in the carbon markets of the United States, Europe, Japan, and China for the given year.

Benefit-cost analysis of typical counties

Based on the above analysis, we can obtain the economic development trends and benefit-cost ratios for each country, which can be used to assess whether it is worthwhile to use GGDP as the primary economic indicator on a global scale.

Figure 8 shows the changes in GDP for each country after adopting GGDP as the primary economic indicator, with developed countries on the left and developing countries on the right. Here, $RGDP$ represents the actual GDP of the country, and $OGDP$ represents the optimal GDP when using GGDP as the primary economic indicator. Overall, for both developed and developing countries, the value of $OGDP - RGDP$ is negative after adopting GGDP, meaning that the expected GDP ($OGDP$) initially decreases and is less than $RGDP$. However, the growth rate of $OGDP$ is greater than $RGDP$, and the final value of $OGDP$ approaches or even exceeds $RGDP$.

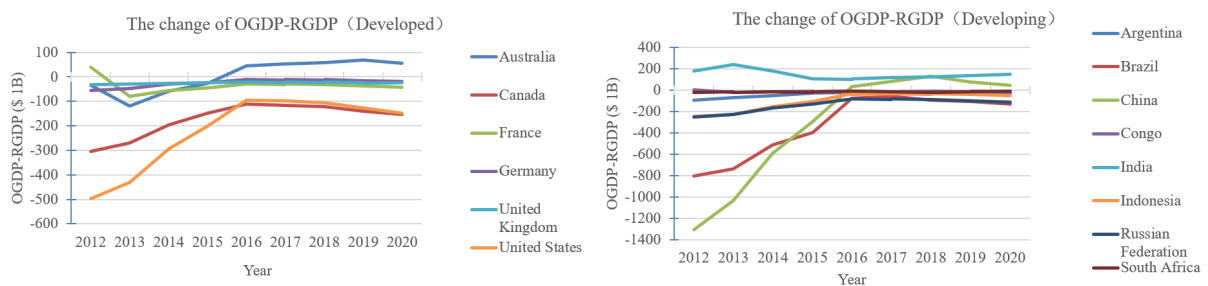


Figure 8: Change of GDP

The trend of GGDP changes in each country is shown in Figure 9. The left side shows developed countries, while the right side shows developing countries. It can be observed from

the figure that developed countries, on the whole, experience an increase in GGDP after adopting GGDP, indicating that they attach great importance to ecological environmental protection. However, developing countries exhibit two distinct trends. China and India, for instance, experience an initial increase and subsequent decrease in the value of $OGGDP - RGGDP$, followed by another increase. On the other hand, countries such as South Africa experience a slow increase in the value of $OGGDP - RGGDP$ around 0. This is probably because developed countries are more likely to improve energy efficiency and find a balance between economic and ecological development, whereas less developed countries, such as South Africa, would have to pay a high economic cost to change the status quo, resulting in minimal expected changes after adopting GGDP, with $OGGDP$ approximating $RGGDP$.

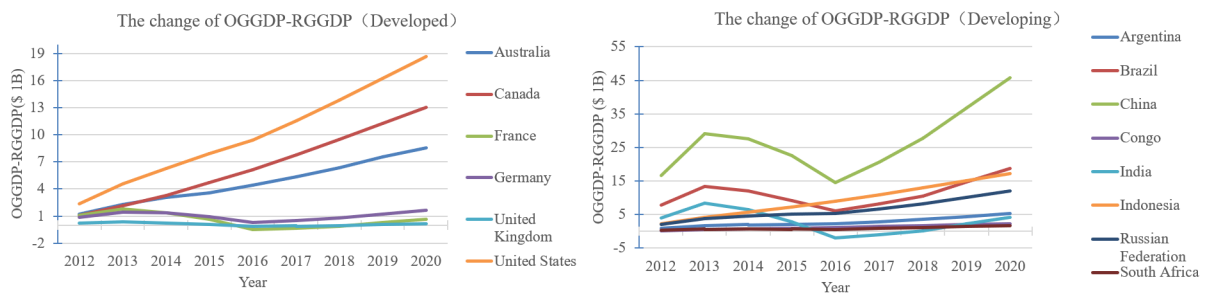


Figure 9: Change of GGDP

Based on the potential benefit and potential cost obtained in the previous section, we calculate the benefit-cost ratio of each country, as shown in Figure 10. If the benefit-cost ratio is greater than 1, it indicates that the potential benefit is greater than the potential cost. The change trend of the benefit-cost ratio is more obvious in developed countries, with most countries exhibiting an increasing trend. Therefore, in the long run, it is expected that in most developed countries, the potential benefit of implementing GGDP will be greater than the potential cost. The change trend of the benefit-cost ratio is relatively unstable in developing countries, but it still shows an overall increasing trend.

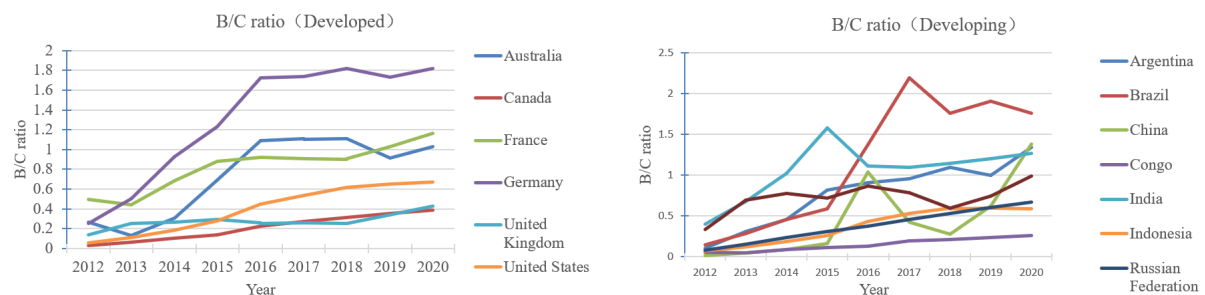


Figure 10: Change of benefit-cost ratio

In the long run, regardless of which kind of country, the potential benefit may be greater than the cost. Therefore, we believe that after adopting GGDP globally, countries will make efforts to protect the environment, and this effort will be worthwhile. In the short term, GDP growth in each country may be affected to a certain extent, but in the long run, the growth rate of GDP will gradually increase and eventually approach or even exceed the actual value based on GDP. Developed countries may be able to make adjustments more quickly.

7 Deep Analysis in China

We select China to develop a more in-depth analysis, because China's significant production of global CO₂e emissions and its large and diverse economy.

7.1 Use of Natural Resources

Firstly, we analyze the changes in China's use of natural resources by calculating relevant variables using GGDP, as well as the actual index based on economic data. We obtain the expected area of different types of ecosystems and the actual area of ecosystems. Figure 11 shows the changes in the natural resource area of land. We find that when using GGDP as the primary economic indicator, the areas of lakes, wetlands, and forests are all larger than when using GGDP as the primary economic indicator. This indicates that the government recognizes the importance of environmental protection in pursuit of maximizing GGDP, and is no longer blindly pursuing rapid economic growth at the expense of the environment, which is consistent with our theme of sustainable development. At the same time, it can be observed that the areas of cropland and grassland are both smaller when using GGDP as the primary economic indicator, which indicates that using GGDP as a measure will inevitably cause some economic losses. However, the short-term cost is worthwhile, and it will lead to long-term stable development of the economy and human society. It is also noteworthy that even when using GGDP as the primary economic indicator, there is still a significant decrease in the area of wetlands, which is due to the high cost of wetland conservation and the slow recovery of the environmental carrying capacity of wetlands.

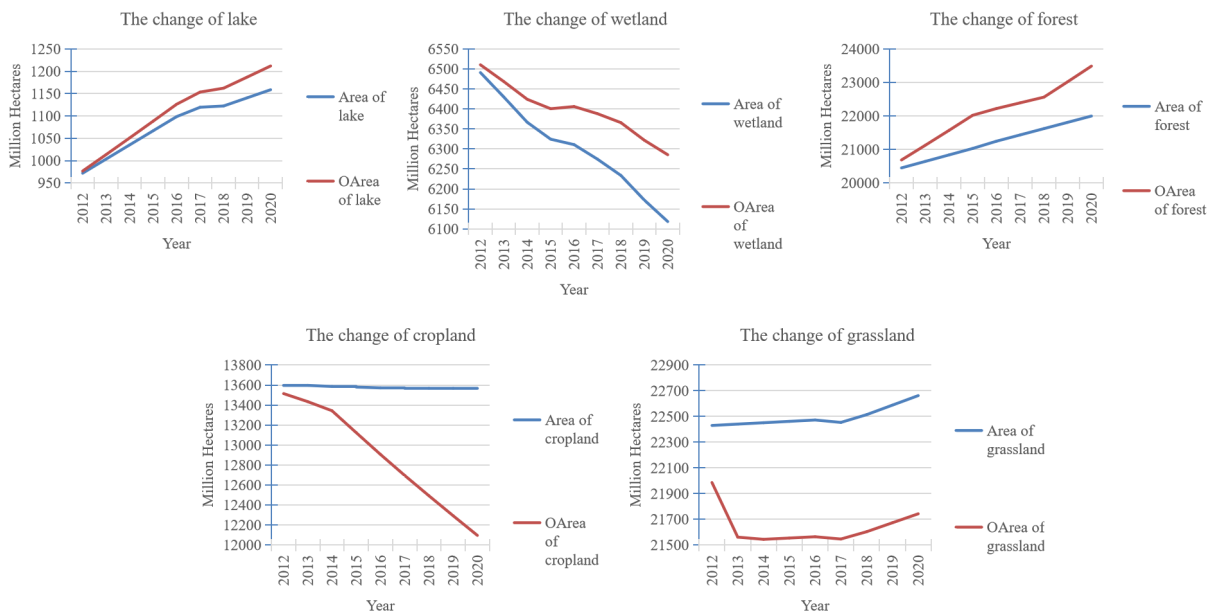


Figure 11: Change of lands

7.2 Net Benefit

To better measure the changes in a country's development level after using GGDP, we calculate the net benefit of adopting GGDP, defined as follows:

$$NetBenefit = AI - EC$$

Where AI represents the improvement of the ability to support future generations, and EC represents the economic cost. If the country's economic status declines, then EC is positive.

The calculation of AI is as follows:

$$AI = Oability - Rability \quad (18)$$

$$Oability = \sum_{k=1}^n OA_k \times TV_k \quad (19)$$

$$Rability = \sum_{k=1}^n RA_k \times TV_k \quad (20)$$

Where $Oability$ represents the total ecological service value of the country's ecosystems after adopting GGDP, and $Rability$ represents the same value under the current GDP system. The expected area of the k -th ecosystem in the country after adopting GGDP as the primary economic index is denoted as OA_k , and the actual area is denoted as RA_k . TV_k represent the total ecosystem service value of per unit area of k -th ecosystem. Natural resources are the foundation of future generations' development, and all human development relies on them. Therefore, we use the "unit value" recorded in figure 2 to measure the potential long-term value of each type of ecosystem, by including all 17 values listed in figure 2, to evaluate a country's ability to support future generations.

EC represents the economic cost, calculated as:

$$EC = RGDP - OGDP \quad (21)$$

Assuming other countries still use GDP as their primary economic index, China is required to use GGDP for certain reasons. Therefore, we use the change in GDP to measure the changes in economic status. $OGDP$ represents the expected GDP after using GGDP, and $RGDP$ represents the actual GDP.

The trend of net benefit changes is shown in Figure 12. We assume that in the first year of implementing GGDP (2012), China's net benefit is negative. It quickly becomes positive after 4 years and remains stable thereafter. The results indicate that although net benefit is negative in the short term after adopting GGDP, it is positive in the long term. Thus, the implementation of GGDP is beneficial for China's development in the long run.

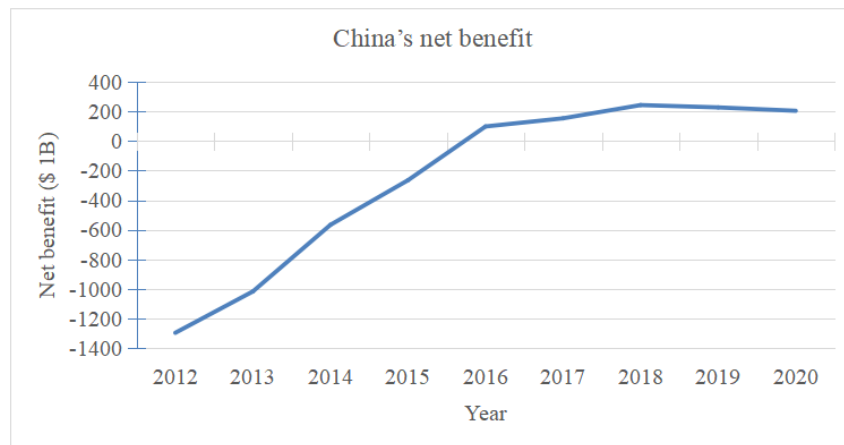


Figure 12: China's net benefit

7.3 Change of Economic Status

We conducted an in-depth analysis of the change in economic status resulting from the implementation of Green GDP (GGDP). Figure 13 displays the possible changes in Gross Domestic Product (GDP) and GGDP before and after their implementation in China. First, we examine the changes in GDP. After adopting GGDP, China's GDP initially decreased compared to actual GDP (RGDP), but gradually converged to its value. On the other hand, optimal GGDP (OGGDP) was greater than actual GGDP (RGGDP), and its growth trend was higher than that of actual GGDP(RGGDP).

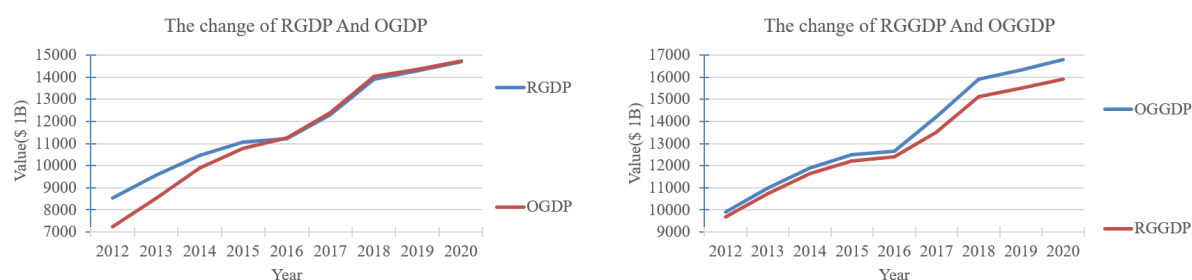


Figure 13: Change of economic status

7.4 Improvement of Ability to Support Generation

The figure depicted in Figure 14 illustrates the changes in China's ability to support future generations before and after the implementation of green GDP. It is evident that the adoption of GGDP has led to a significant improvement in the ability to support future generations.

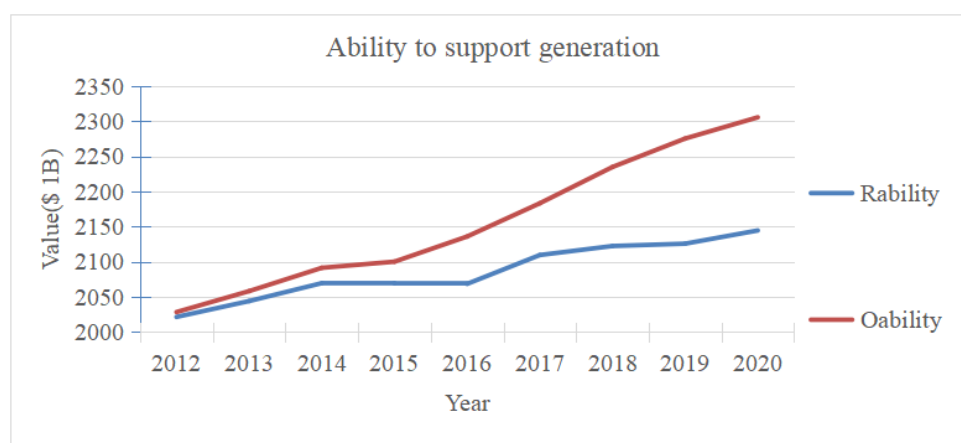


Figure 14: Change of ability to support generation

These results suggest that in the long term, economic development can at least remain the same as before, while the ecological environment can be improved obviously, thereby enhancing China's sustainable development capacity.

7.5 Sensitivity Analysis

The degree to which a country values one unit of CO₂e emissions is closely related to the measurement of GGDP. This degree of importance can be reflected by the country's pricing of

one unit of CO₂e. Therefore, we can conduct sensitivity analysis by varying this indicator. We adjusted the unit price of CO₂e by 10% based on the original data and obtained the following results:

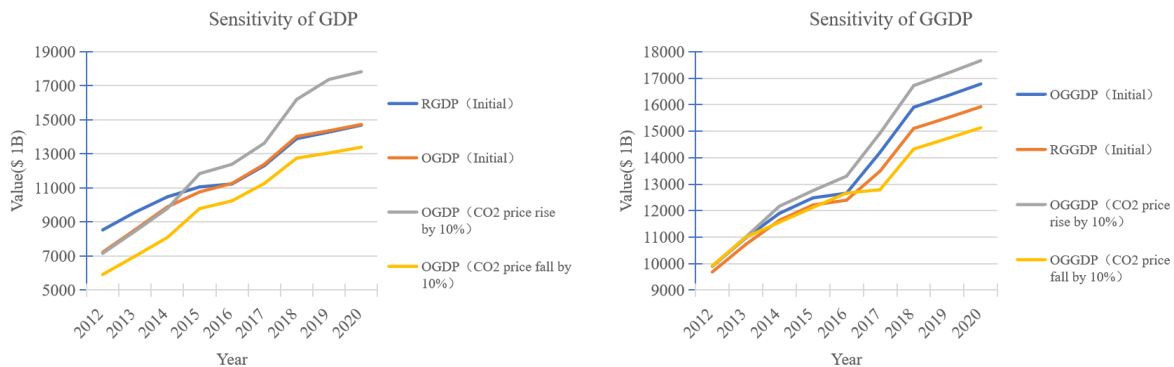


Figure 15: Sensitivity Analysis

From the results, it can be seen that even after adjusting the carbon price up or down by 10%, the conclusions drawn in the previous analysis are still supported. This demonstrates that our model has good robustness.

8 Evaluation of Models

8.1 Strength

- When evaluating Green GDP, we consider both the negative impacts of human activities such as pollution and environmental damage, and the value of natural resources in providing ecosystem services to humans. However, many studies that measure Green GDP simply add the traditional GDP and the value of a nation's ecosystem services without distinguishing the portion that can be considered as a final product. We have identified this portion.
- Based on reasonable assumptions, we have developed a non-linear programming model to examine the trade-off between economic development and ecological sustainability after implementing Green GDP. We assume that a country's objective is to maximize its Green GDP.
- We have selected 14 countries worldwide as representative countries to calculate the impact of implementing Green GDP on climate improvement, which can be generalized to the global scale. These countries include both developed and developing countries, with a mix of well-developed and less-developed economies. Therefore, these countries are typical and the results of our analysis can be highly credible.
- We have considered all greenhouse gases and converted the data measured in CH₄ and N₂O, as well as other greenhouse gases, to carbon dioxide equivalent using Global Warming Potential (GWP), which provides a unified measurement approach and a more comprehensive assessment of the climate impact.

8.2 Weaknesses

- The incomplete statistics on natural resources in some countries around the world make it difficult to accurately measure the impact of using Green GDP as an alternative indicator on global climate.
- Additionally, we may not be able to accurately predict a country's decision-making after adopting GGDP as the primary economic indicator.

9 Conclusion

We incorporated the value of the natural environment into the traditional GDP and deducted the damages caused by human activities to derive the green GDP (GGDP). GGDP not only measures economic development and national strength but also reflects the country's sustainable development capacity through natural resource accounting. Using a nonlinear programming model, we identified the optimal solution and maximum value of GGDP and compared it with the traditional GDP. We also calculated the potential benefits and costs of adopting GGDP for various typical countries and extended the findings to the global level.

As a case study, we examined China, a typical country, and found that adopting GGDP as the primary economic indicator can lead to higher GDP growth rates in the long term compared to using traditional GDP. Furthermore, we analyzed various types of land and identified the substantial value of forests and wetlands to the global environment, emphasizing the need for their protection. Ultimately, we provided recommendations to Chinese leaders to adopt GGDP as a measure of national strength and promote sustainable development.

Non-technical Report

To: leaders of China

From: Team 2309944

Date: February 21st, 2023

Subject: Recommendation for China to Adopt Green GDP for Long-Term Sustainability

We are writing to recommend the Chinese government to adopt Green GDP (GGDP) as the primary measure of national economic health, which can contribute to the long-term sustainability of the country.

In today's society, it is undeniable that many countries prioritize economic development at the expense of the environment, leading to the greenhouse effect. As a result, it has become increasingly apparent that traditional economic metrics like Gross Domestic Product (GDP) fail to capture the environmental effects of economic activities adequately. As a response, many countries have started exploring alternative measures of economic well-being, such as Green GDP. This metric takes into account the negative externality of economic activities, including their environmental impact, providing a more comprehensive view of the real costs and benefits of economic activity. By doing so, it offers a more comprehensive perspective on the true costs and benefits of economic activity. This can help promote sustainable economic growth and investment in low-carbon technologies, while discouraging activities that have significant negative environmental impacts.

As one of the world's largest economies, China has a significant role to play in addressing the climate crisis. According to our team's calculations, using GGDP as the primary economic indicator would result in a higher GDP growth rate in the long run compared to using traditional GDP. We find that in the long term, economic development can at least remain the same as before, and the ecological environment can be improved obviously, thereby enhancing China's sustainable development capacity.

Furthermore, to facilitate the calculation of GGDP, it is important for the statistical agencies that collect and maintain data related to the environment and natural resources to increase their efforts and share their data. This will provide a more accurate and comprehensive view of the country's economic performance and environmental impact.

In conclusion, we believe that adopting GGDP as the primary measure of economic health in China is essential for achieving sustainable development and its ability to support future generations. We hope that our recommendations will be seriously considered by the government as it makes important policy decisions for the country's future.

References

- [1] World Development Indicators | DataBank [worldbank.org](https://data.worldbank.org/)
- [2] Country Climate and Development Report (CCDR) | DataBank [worldbank.org](https://data.worldbank.org/)
- [3] Global Wetland Outlook 2018 <https://www.global-wetland-outlook.ramsar.org/gwo-2018>
- [4] The Food and Agriculture Organization (FAO) <https://www.fao.org/faostat/en/data>
- [5] Lu Songhan. Integrated evaluation scale for regional development: from GDP, GGDP to MDP[J]. Science Popularization, 2007, No.488(02):100-101.
- [6] Jin Chunxiang, Ni Wei, Ni Baichun. A preliminary analysis of "GDP" and "Green GDP"[J]. Forestry Survey and Design, 2009, No.151(03):43-45.
- [7] Lei Ming. Sustainable development evaluation—accounting for green GDP (GGDP)[J]. Economic Research Reference, 2001, (29):14-24.
- [8] Ye Feifei, Yang Longhao, Wang Yingming. Prediction method of environmental governance cost considering input-output relationship and efficiency[J]. Control and Decision, 2020, 35(04):993-1003.
- [9] Boyd, J. Nonmarket benefits of nature: What should be counted in green GDP? Ecological Economics, 2007, 61(4), 716–723.
- [10] Callan, T. Gross Domestic Product: An Economy's All. International Monetary Fund, Economics Concepts Explained, 2023.
- [11] Costanza, R., de Groot, R., Sutton, P., van der Ploeg, S., Anderson, S. J., Kubiszewski, I., Farber, S., Turner, R. K. Changes in the global value of ecosystem services. Global Environmental Change, 2014, 26, 152–158.
- [12] Costanza R, d'a'Arge R, de Groot R, et al. The value of the world's ecosystem services and natural capital. Nature, 1997, 387:235-260.
- [13] Gao W. S, Chen Y. Q., Chen D. D. A New Index System of Agricultural Green Accounting Based on Value of Agroecosystem Services. Research of Agricultural Modernization, 2005, 26(1):1-5.
- [14] Li, V.; Lang, G. China's 'Green GDP' Experiment and the Struggle for Ecological Modernisation. Journal of Contemporary Asia, 2010, Vol. 40, No. 1, pp. 44-62.
- [15] Piao, S., Liu, Q., Chen, A., Janssens, I. A., Fu, Y., Dai, J., Liu, L., Lian, X., Shen, M., Zhu, X. Plant phenology and global climate change: Current progresses and challenges. Global Change Biology, 2019, 25(6), 1922–1940.
- [16] Stjepanović, S., Tomić, D., Škare, M. A new approach to measuring green GDP: A cross-country analysis. Entrepreneurship and Sustainability Issues, 2017, 4(4), 574–590.
- [17] Tan, Y., Liu, X., Sun, H., Zeng, C. Population ageing, labour market rigidity and corporate innovation: Evidence from China. Research Policy, 2022, 51(2), 104428.
- [18] Xu, L., Yu, B., Yue, W. A method of green GDP accounting based on eco-service and a case study of Wuyishan, China. Procedia Environmental Sciences, 2010, 2, 1865–1872.